2. MOLECULES AND ATOMS

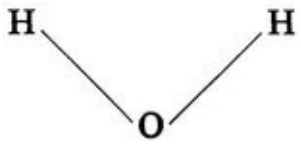
I. MOLECULAR STRUCTURE

In order to prepare the way for the actual subject of this book, the second lecture will deal first with the structure of molecules, and then with the structure of atoms.

Visualize a piece of silver. You can cut it up, first, with coarse mechanical tools, and then with a file you can reduce these fragments to almost invisibly minute specks of dust. But neither of these operations has enabled you to reach the smallest component particles. If you rub the silver with your hand, a minute quantity of the metal will stick to your hand. But even this hardly at all visible quantity contains an enormous number of silver atoms. Finally, you can heat the silver until it melts and finally evaporates, in other words, until it becomes a gas. This process breaks it up into its ultimate, smallest particles, the atoms. At any rate, it cannot be broken up further either mechanically or by chemical means. Silver is a pure element.

But if a drop of water is caused to vaporize, it is not broken down into water atoms. The smallest particles which can be obtained in this manner, the water molecules, can still be broken down further by chemical means. A water molecule consists of two hydrogen atoms and one oxygen atom. Water is not an element.

Modern physics has established a definite geometrical picture of the structure of such a molecule. The hydrogen atom is denoted by the symbol H, the oxygen atom by the symbol O. With these symbols, the water molecule can be represented by the following schematic formula:



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For certain reasons, which cannot be discussed further here, the water molecule is imagined as the triangular structure shown diagrammatically in Figure 1. In this sketch, shading indicates the average distribution of the electric charge of the atoms.

Now, proceeding from the individual molecule to matter in its various states of aggregation—for instance, from the water molecule to water vapour, water and ice—we can extend this visual presentation as follows: In water